BY RAKA ARRAYAN

EMAIL CAMPAIGN ANALYSIS

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SOMETIMES COMPLICATED MATH HAS A SIMPLE ANSWER

BACKGROUND

IN TODAY'S DIGITAL ERA, SMALL TO MEDIUM BUSINESS OWNERS ARE INCREASINGLY UTILIZING EMAIL-BASED MARKETING STRATEGIES VIA GMAIL TO CONVERT POTENTIAL CUSTOMERS INTO LOYAL CUSTOMERS. MARKETING EMAILS ALLOW BUSINESSES TO CHARACTERIZE AND TRACK INTERACTIONS, SUCH AS WHETHER EMAILS ARE IGNORED, READ, OR ACKNOWLEDGED BY RECIPIENTS, WHICH HELPS MEASURE CAMPAIGN EFFECTIVENESS. INSPIRED BY THE PRINCIPLE THAT ADVERTISING EXPENDITURE IS DIRECTLY PROPORTIONAL TO SALES, REGRESSION ANALYSIS CAN BE USED TO PRECISELY QUANTIFY THIS RELATIONSHIP. WE ENCOURAGE EVERYONE TO EXPERIMENT WITH THIS DATA AND EXPLORE THE VALUE OF EMAIL AS A TOOL IN A MULTI-CHANNEL MARKETING STRATEGY FOR SMALL TO MEDIUM BUSINESSES.

GOALS

THIS DATASET AIMS TO ANALYZE AND IMPROVE THE EFFECTIVENESS OF EMAIL CAMPAIGNS BY STUDYING VARIOUS FACTORS THAT INFLUENCE WHETHER EMAILS ARE OPENED OR NOT BY CUSTOMERS. THE ANALYSIS INCLUDES ELEMENTS SUCH AS EMAIL TYPE, SUBJECT HOTNESS SCORE, EMAIL SOURCE, CUSTOMER LOCATION, CAMPAIGN TYPE, TOTAL_PAST_COMMUNICATIONS, TIME OF EMAIL SENDING, TIME_EMAIL_SENT_CATEGORY, WORD_COUNT,TOTAL_LINKS AND IMAGES,EMAIL STATUS. BY ANALYZING THIS DATA, COMPANIES CAN GAIN USEFUL INSIGHTS TO DESIGN MORE ENGAGING EMAIL CAMPAIGNS, OPTIMIZE SEND TIMES, AND PERSONALIZE EMAILS BASED ON CUSTOMER PREFERENCES, THEREBY INCREASING OPEN RATES AND CUSTOMER ENGAGEMENT.

TOOLS

Google colab is used as platform for analyzing data

Google Sheets is used to make data easier preprocessing is like removing duplicates

pyhton is used data for data analysis processes

ANALYSIS PROCESS

SOMETIMES COMPLICATED MATH HAS A SIMPLE ANSWER

- DATA PREPARATION
- EXPLORATORY DATA ANALYSIS
- PROCESSING DATA
- DATA INSHIGT
- CORRELATION
- MODELING DATA

DATA PREPARATION

DATA IS TAKEN FROM KAGGLE WHERE MANY SMALL TO MEDIUM BUSINESS OWNERS ARE EFFECTIVELY LEVERAGING GMAIL-BASED EMAIL MARKETING STRATEGIES TO CONVERT PROSPECTIVE CUSTOMERS INTO LEADS, THUS MAINTAINING BUSINESS RELATIONSHIPS. TO OPTIMIZE THESE EFFORTS, IT'S CRUCIAL TO ANALYZE VARIOUS ASPECTS OF EMAILS, SUCH AS WHETHER THEY ARE IGNORED, READ, OR ACKNOWLEDGED BY THE RECIPIENT. BY TRACKING THESE INTERACTIONS, BUSINESSES CAN REFINE THEIR APPROACH TO ENSURE MORE SUCCESSFUL ENGAGEMENT WITH THEIR AUDIENCE.

Rows

68353

Columns

12

```
from google.colab import files
upload=files.upload()
```

Choose Files No file chosen Upload widget is

Saving Email_Campaign.csv to Email_Campaign.csv

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 68353 entries, 0 to 68352
Data columns (total 12 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   Email_ID                   68353 non-null  object
 1   Email_Type                 68353 non-null  int64
 2   Subject_Hotness_Score      68353 non-null  float64
 3   Email_Source_Type          68353 non-null  int64
 4   Customer_Location          56758 non-null  object
 5   Email_Campaign_Type        68353 non-null  int64
 6   Total_Past_Communications  61528 non-null  float64
 7   Time_Email_sent_Category   68353 non-null  int64
 8   Word_Count                 68353 non-null  int64
 9   Total_Links                66152 non-null  float64
 10  Total_Images               66676 non-null  float64
 11  Email_Status               68353 non-null  int64
dtypes: float64(4), int64(6), object(2)
memory usage: 6.3+ MB
```

Missing Value

```
Email_ID                         0
Email_Type                       0
Subject_Hotness_Score            0
Email_Source_Type                0
Customer_Location            11595
Email_Campaign_Type              0
Total_Past_Communications     6825
Time_Email_sent_Category         0
Word_Count                       0
Total_Links                   2201
Total_Images                  1677
Email_Status                     0
dtype: int64
```

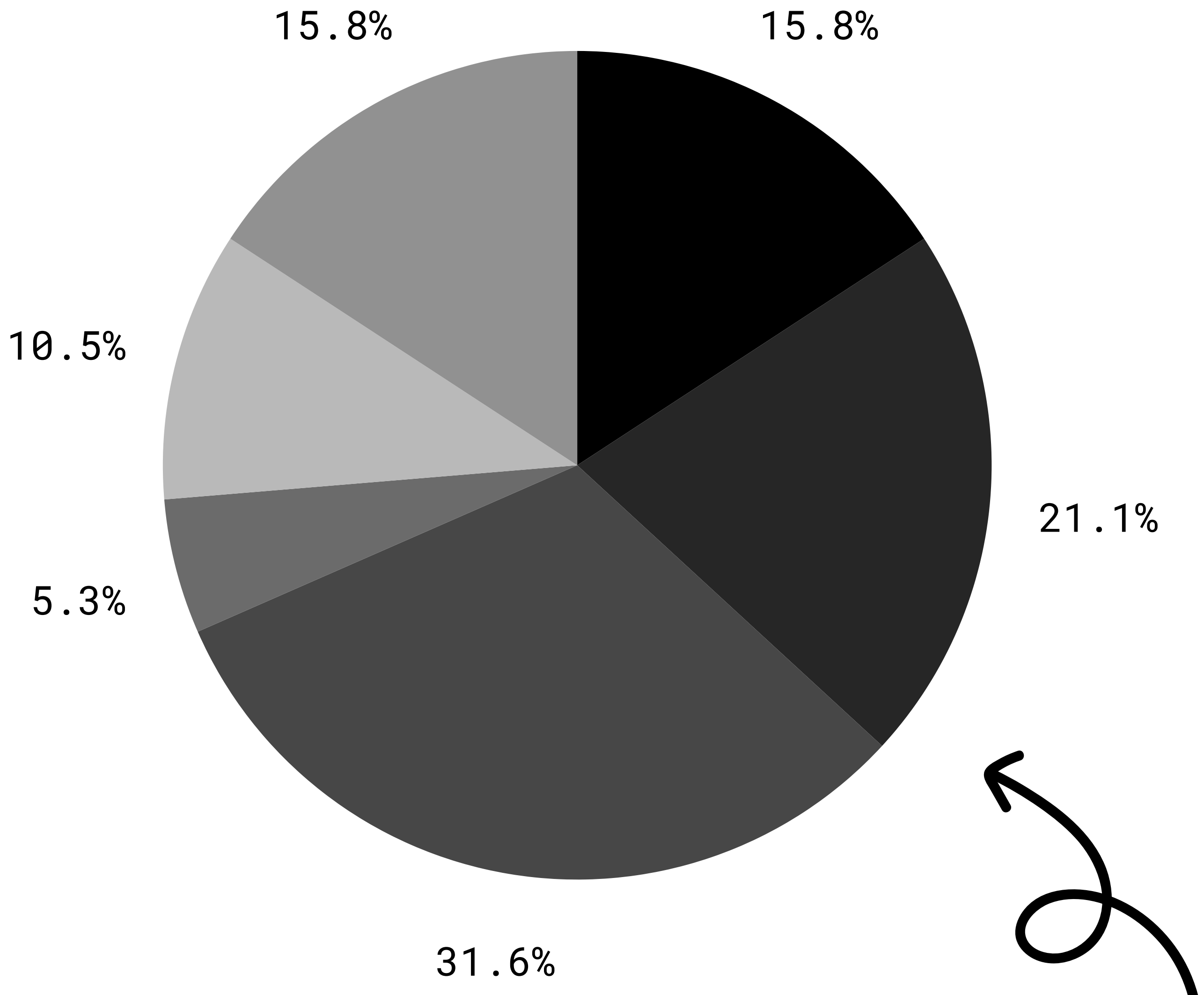
```
# Mengisi missing values pada Customer_Location dengan modus
df['Customer_Location'].fillna(df['Customer_Location'].mode()[0], inplace=True)

# Mengisi missing values pada Total_Past_Communications dengan median
df['Total_Past_Communications'].fillna(df['Total_Past_Communications'].median(), inplace=T

# Mengisi missing values pada Total_Links dengan 0
df['Total_Links'].fillna(0, inplace=True)

# Mengisi missing values pada Total_Images dengan 0
df['Total_Images'].fillna(0, inplace=True)
```

- 18-24
- 25-34
- 35-44

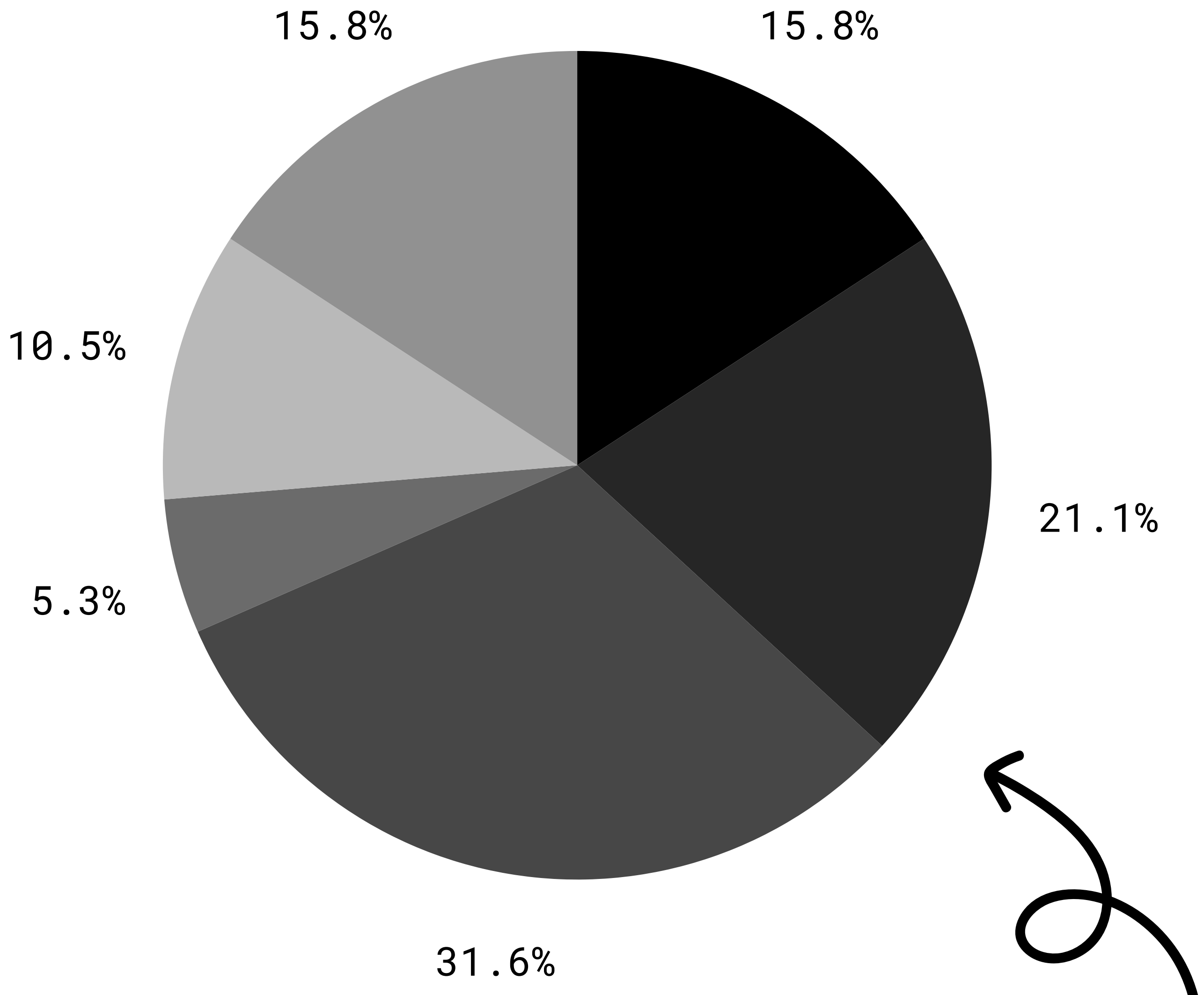


- 45-54
- 55-64
- 65 and up

EXPLORATY DATA ANALYSIS

	Email_Type	Subject_Hotness_Score	Email_Source_Type	Email_Campaign_Type	Total_Past_Communications
count	68353.000000	68353.000000	68353.000000	68353.000000	68353.000000
mean	1.285094	1.095481	1.456513	2.272234	28.840066
std	0.451462	0.997578	0.498109	0.468680	11.897462
min	1.000000	0.000000	1.000000	1.000000	0.000000
25%	1.000000	0.200000	1.000000	2.000000	20.000000
50%	1.000000	0.800000	1.000000	2.000000	28.000000
75%	2.000000	1.800000	2.000000	3.000000	37.000000
max	2.000000	5.000000	2.000000	3.000000	67.000000

Time_Email_sent_Category	Word_Count	Total_Links	Total_Images	Email_Status
68353.000000	68353.000000	68353.00000	68353.000000	68353.000000
1.999298	699.931751	10.09369	3.463564	0.230934
0.631103	271.719440	6.54400	5.555121	0.497032
1.000000	40.000000	0.00000	0.000000	0.000000
2.000000	521.000000	6.00000	0.000000	0.000000
2.000000	694.000000	9.00000	0.000000	0.000000
2.000000	880.000000	14.00000	5.000000	0.000000
3.000000	1316.000000	49.00000	45.000000	2.000000



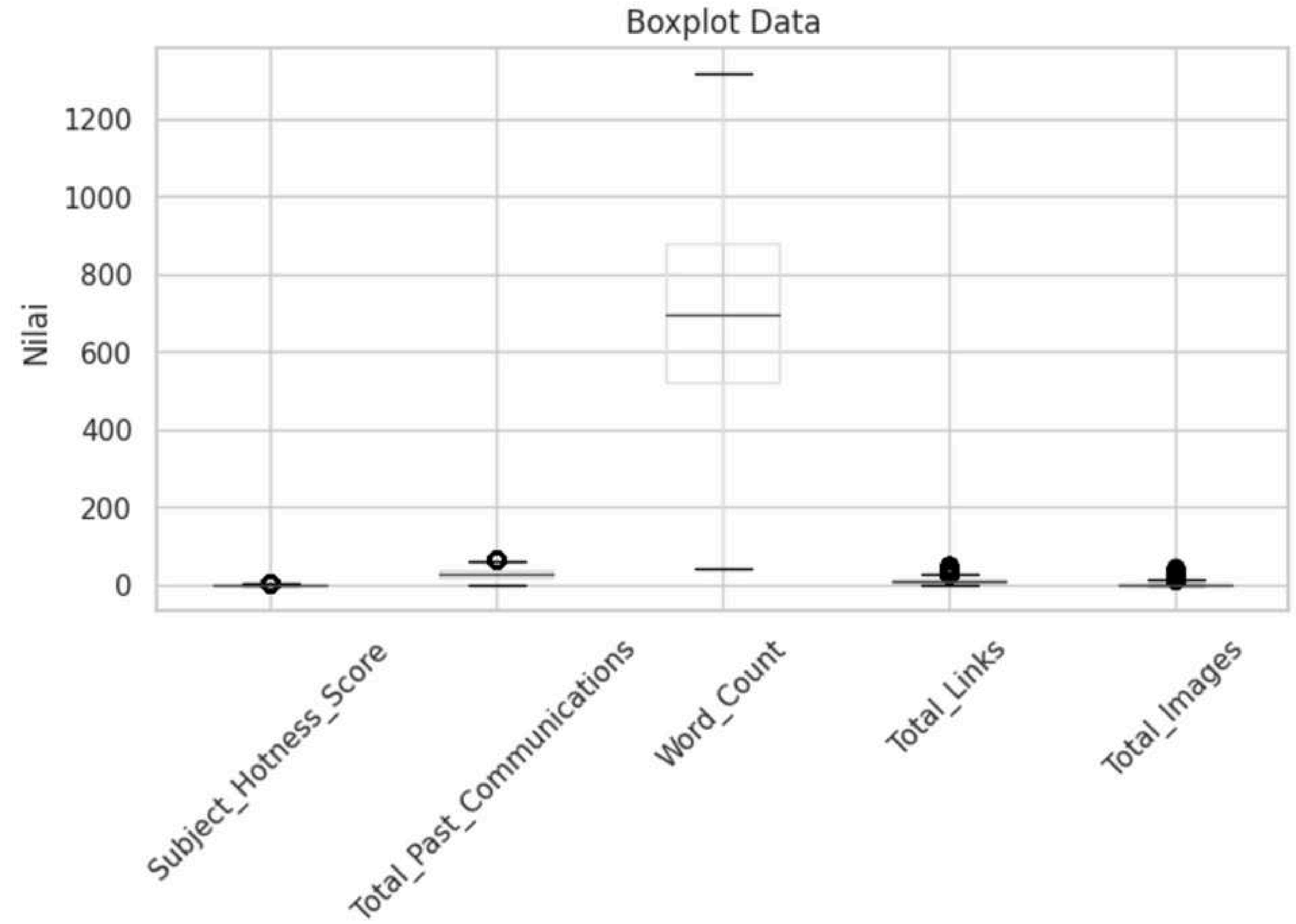
- 18-24
- 25-34
- 35-44
- 45-54
- 55-64
- 65 and up

PROCESSING DATA

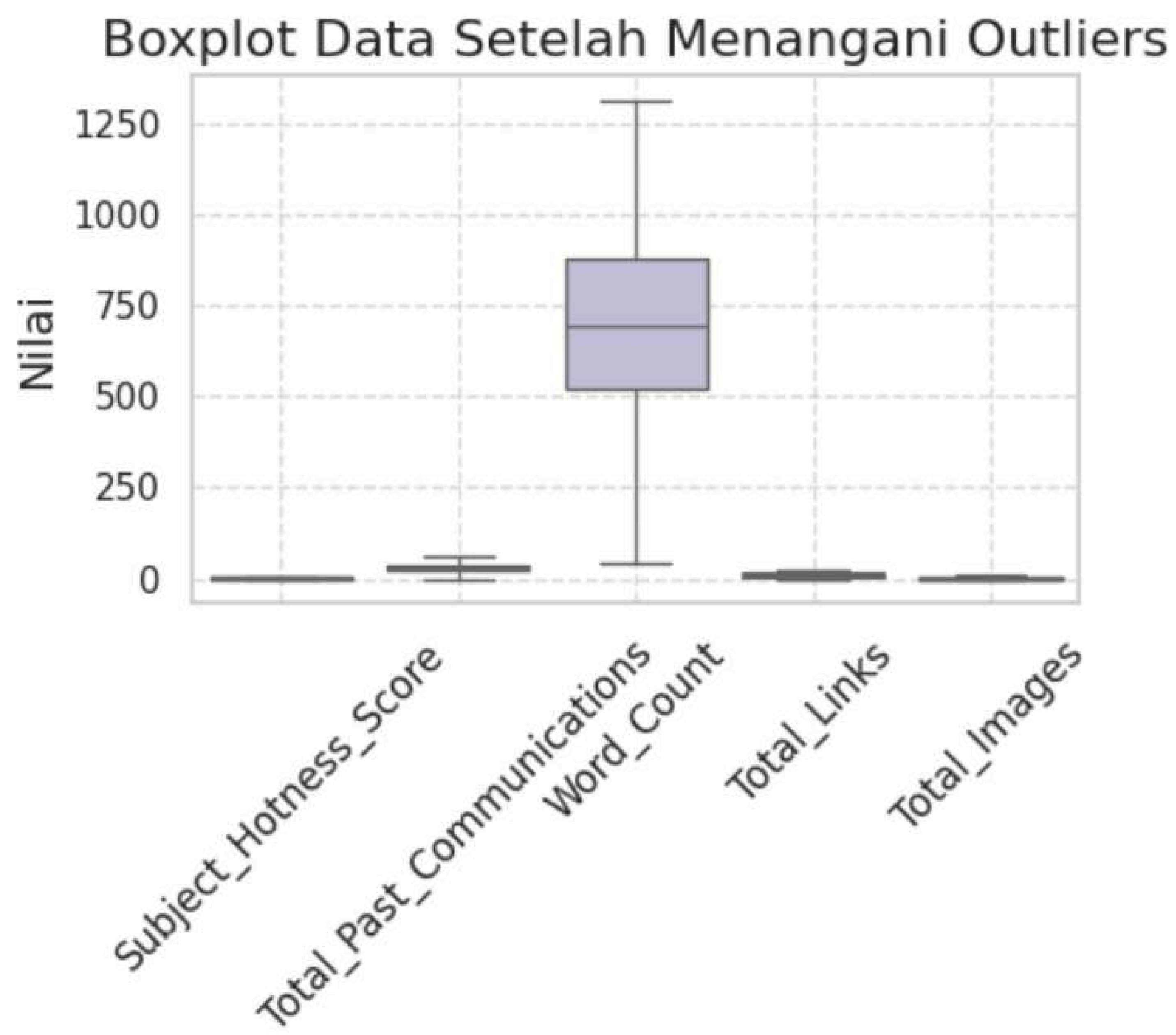
Checking outlier

```
Data Outlier:
Subject_Hotness_Score          247
Total_Past_Communications      136
Word_Count                       0
Total_Links                   1608
Total_Images                  5585
dtype: int64
```

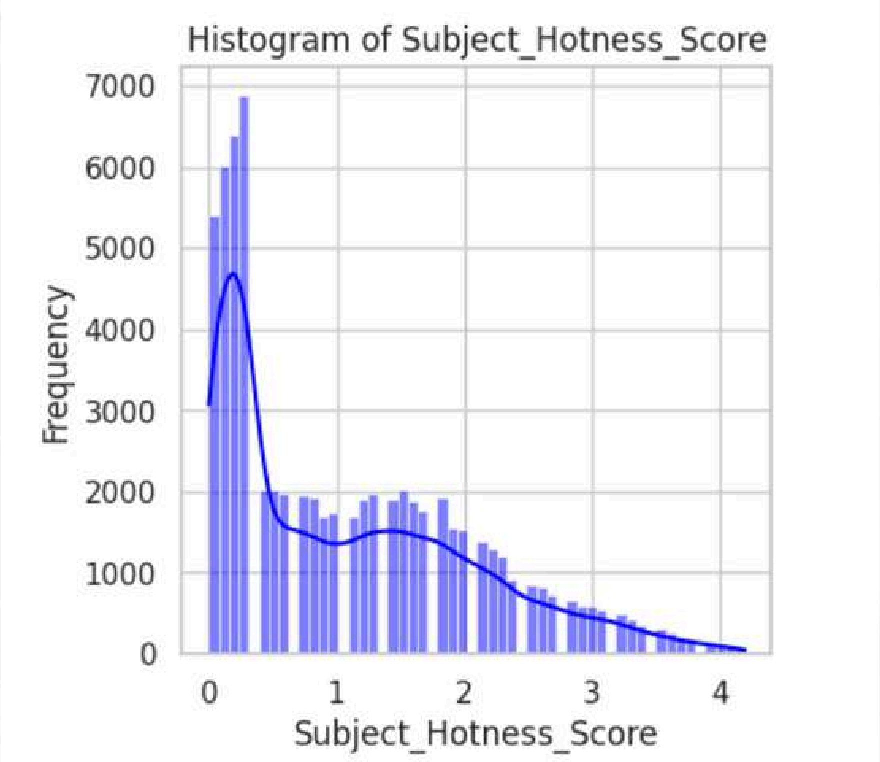
Because there are so many outliers, to reduce the impact of extreme outliers, which can significantly affect the analysis and model results. So we replace the outliers with the values Q1 and Q3

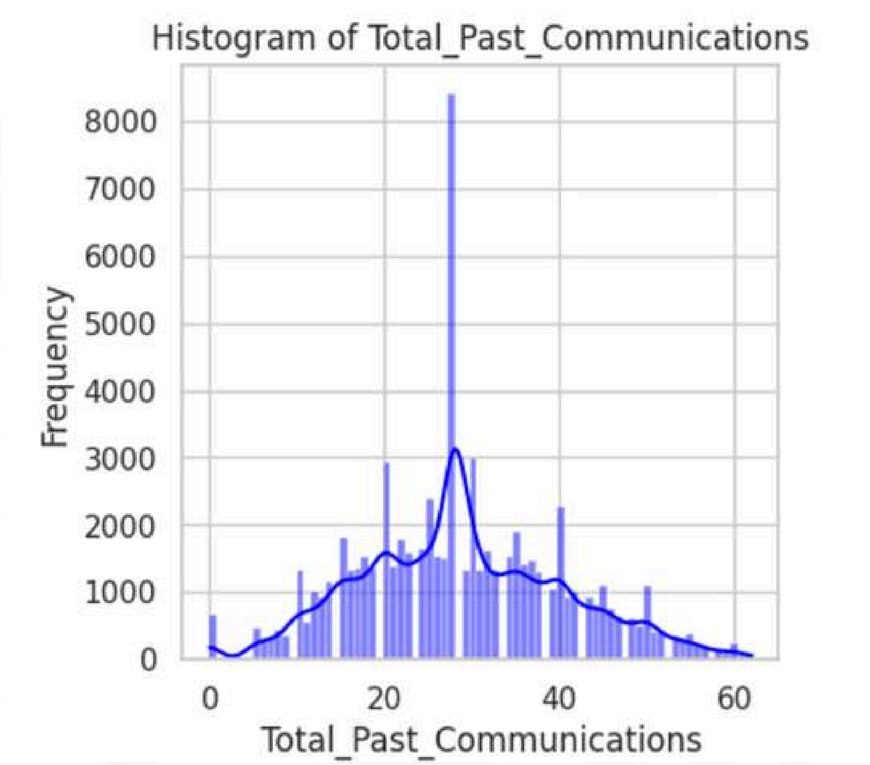


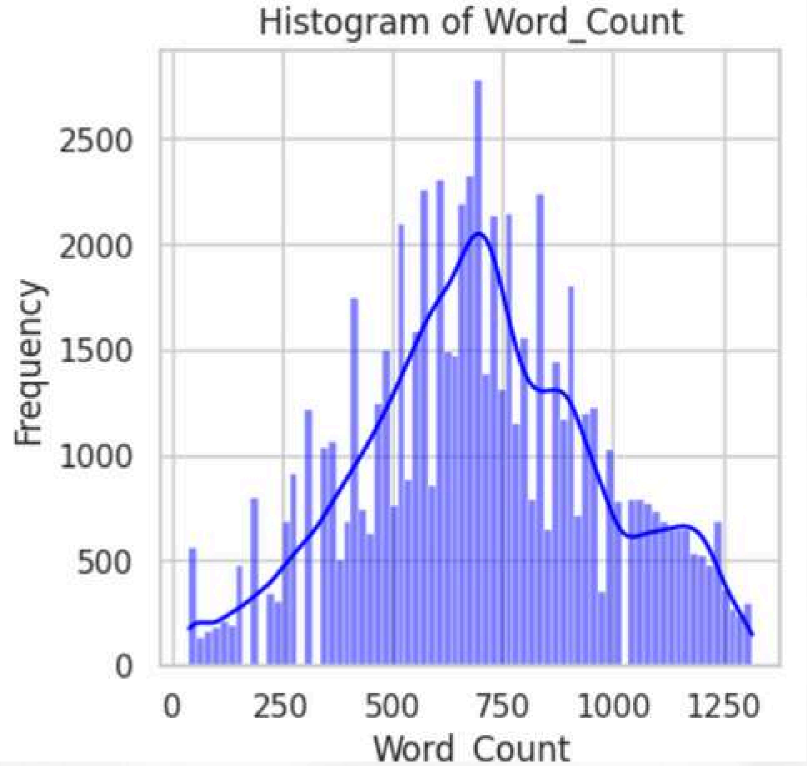
fill in outliers

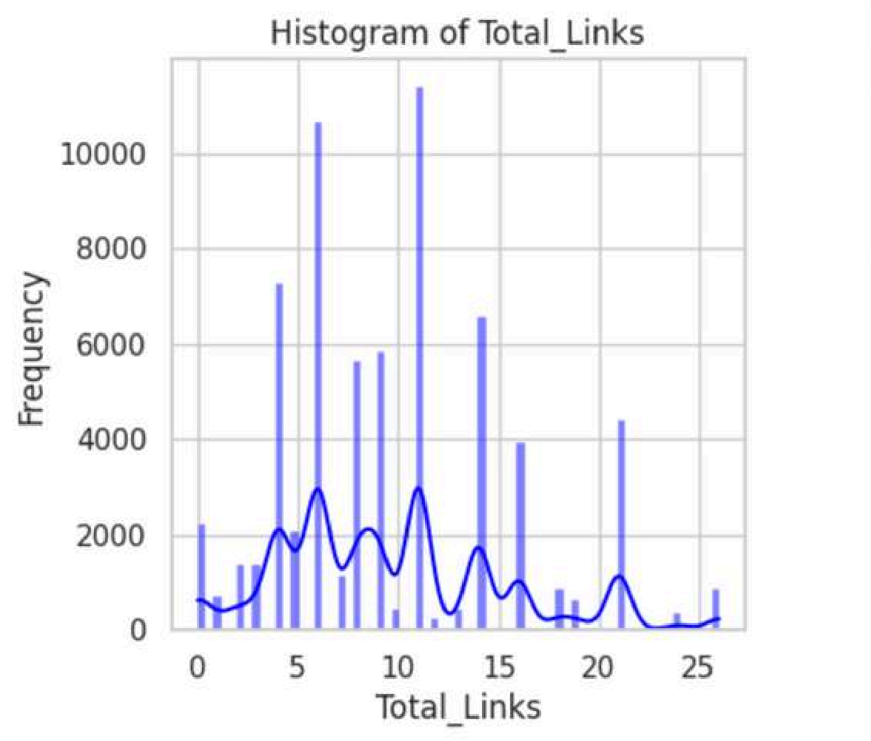


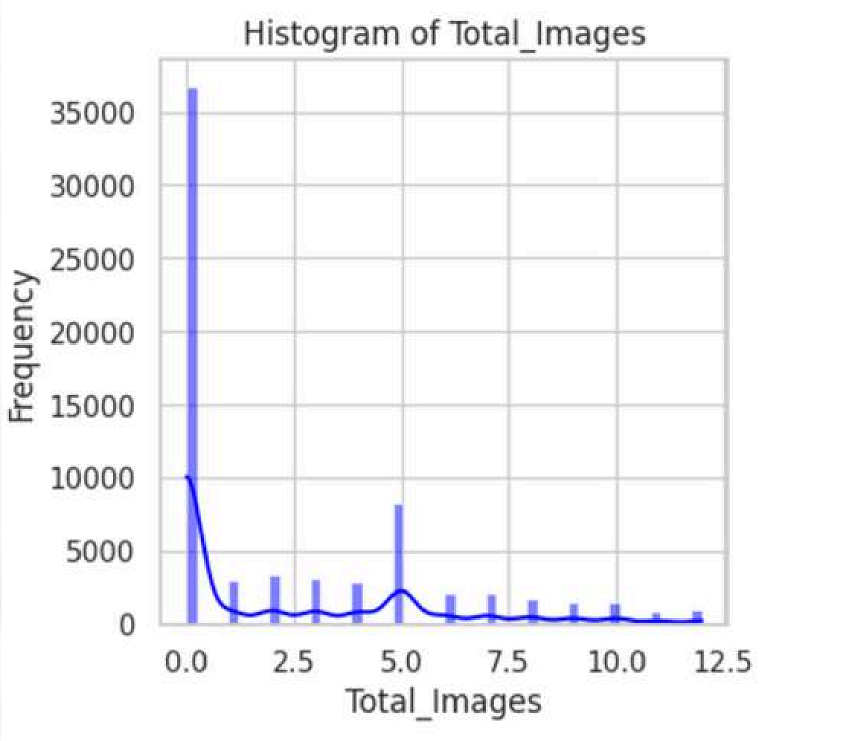
Are there any unreasonable distributions?





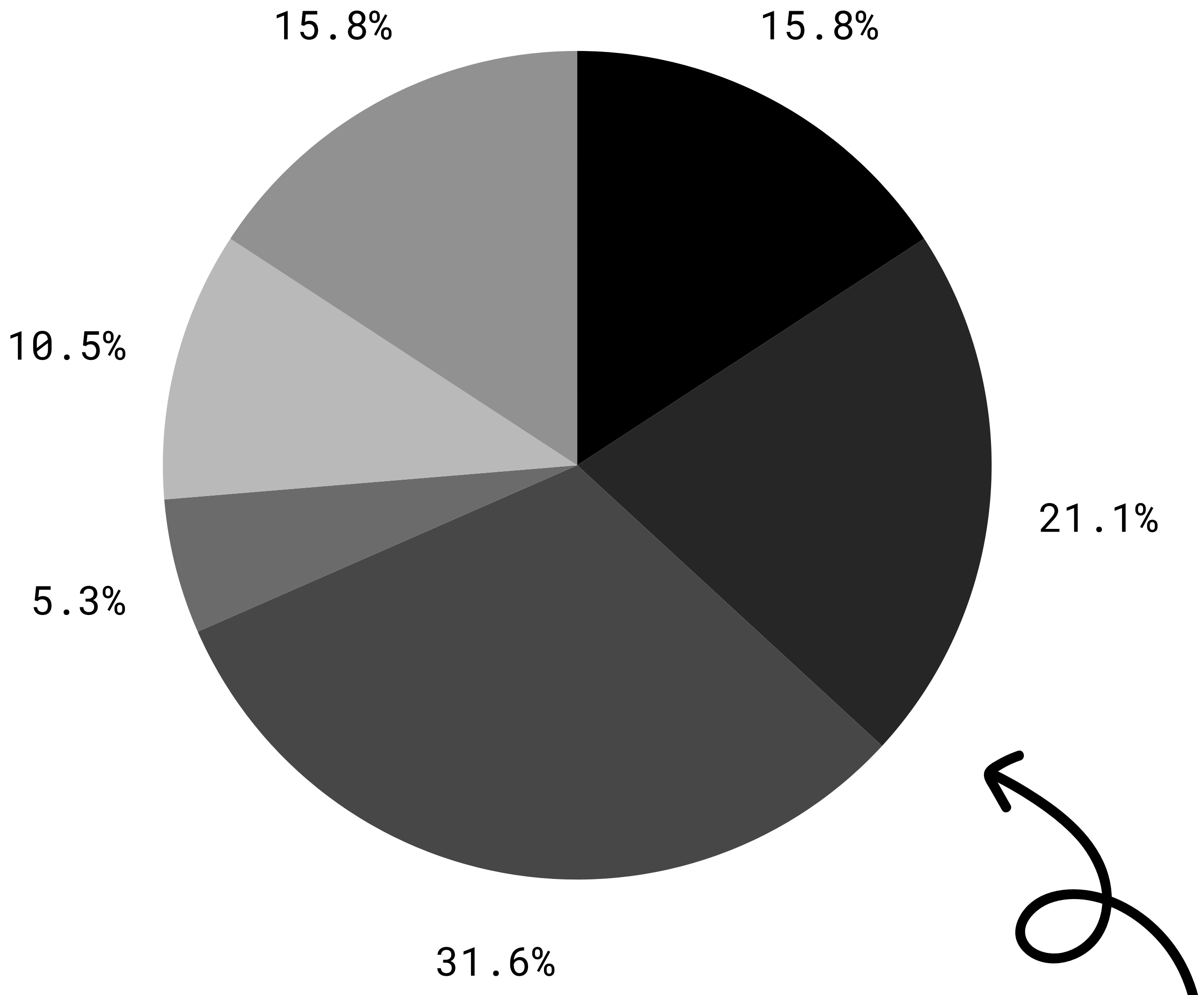






explanation

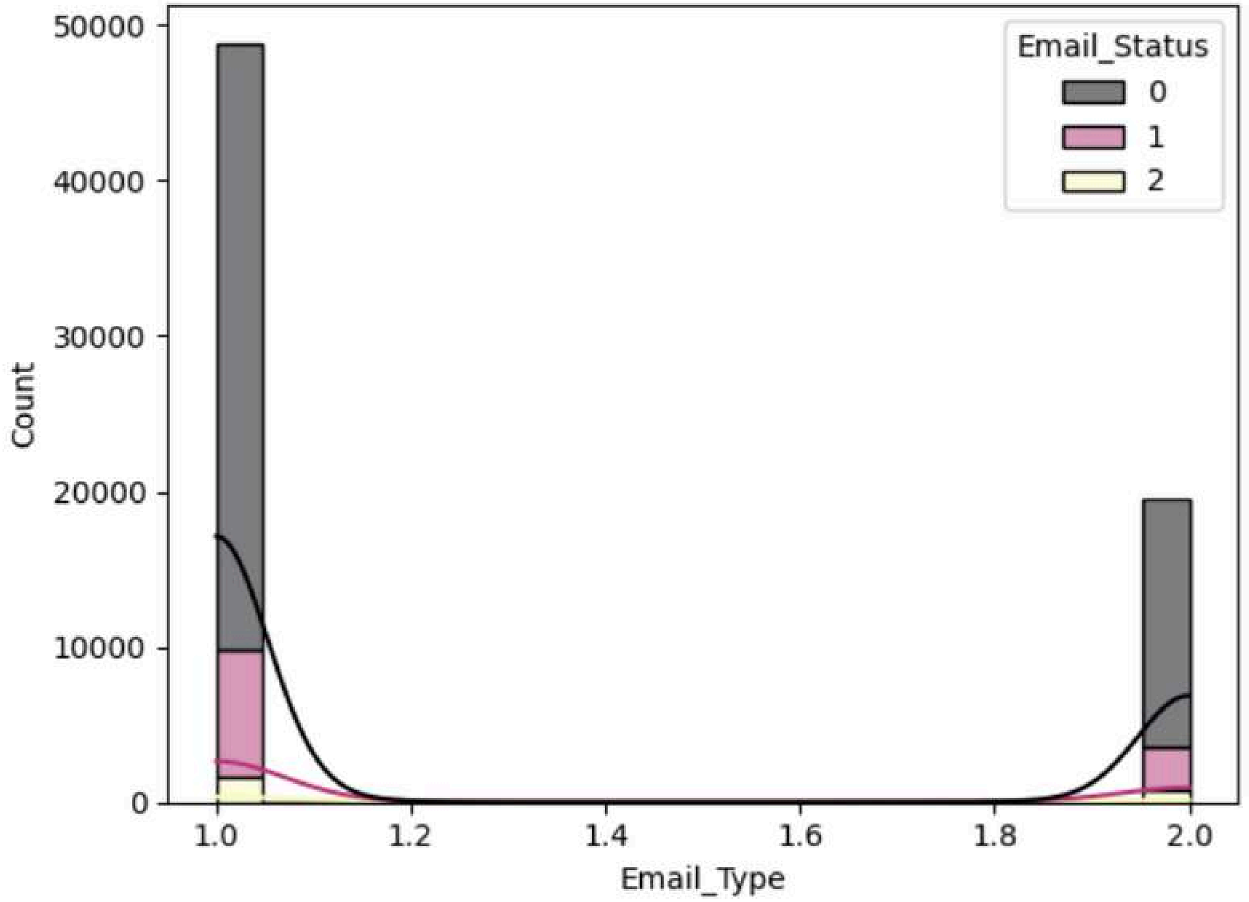
- Most of these distributions show data that is skewed to the right, especially in Subject_Hotness_Score, Total_Past_Communications, Total_Links, and Total_Images, indicating that most values converge in the low values.
- The Word_Count distribution appears to follow a more balanced normal distribution.
- No distribution looks particularly unnatural, but some distributions show that the data is heavily centered on low values, which may be a natural characteristic of the data and indicates that there are rare values in the data.



- 18-24
- 25-34
- 35-44
- 45-54
- 55-64
- 65 and up

DATA INSIGHTS

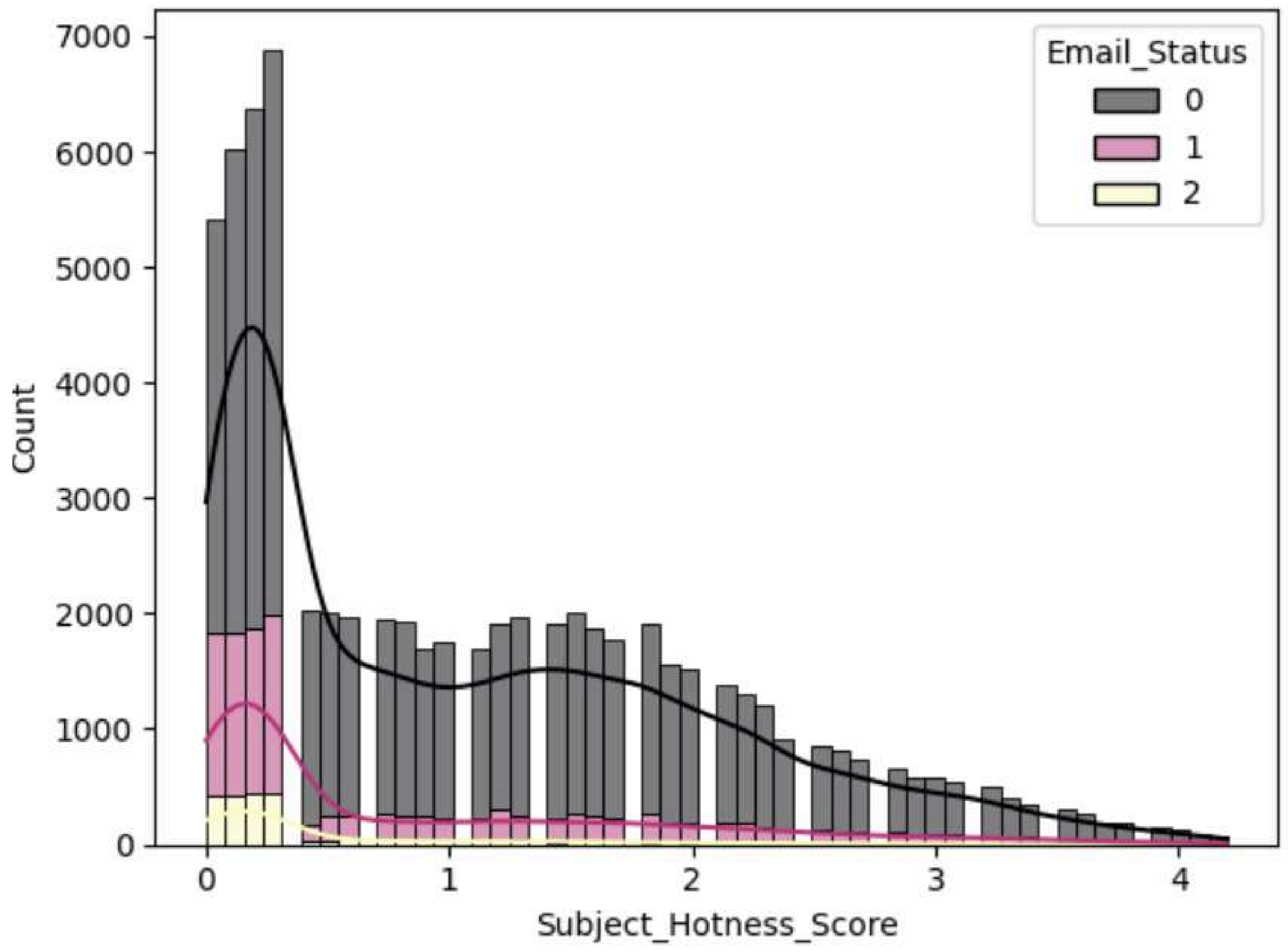
Email Type



Effectiveness of Email Types:
Both types of email (1 and 2) seem to be less effective in attracting recipients' attention, because most of them do not open the email.

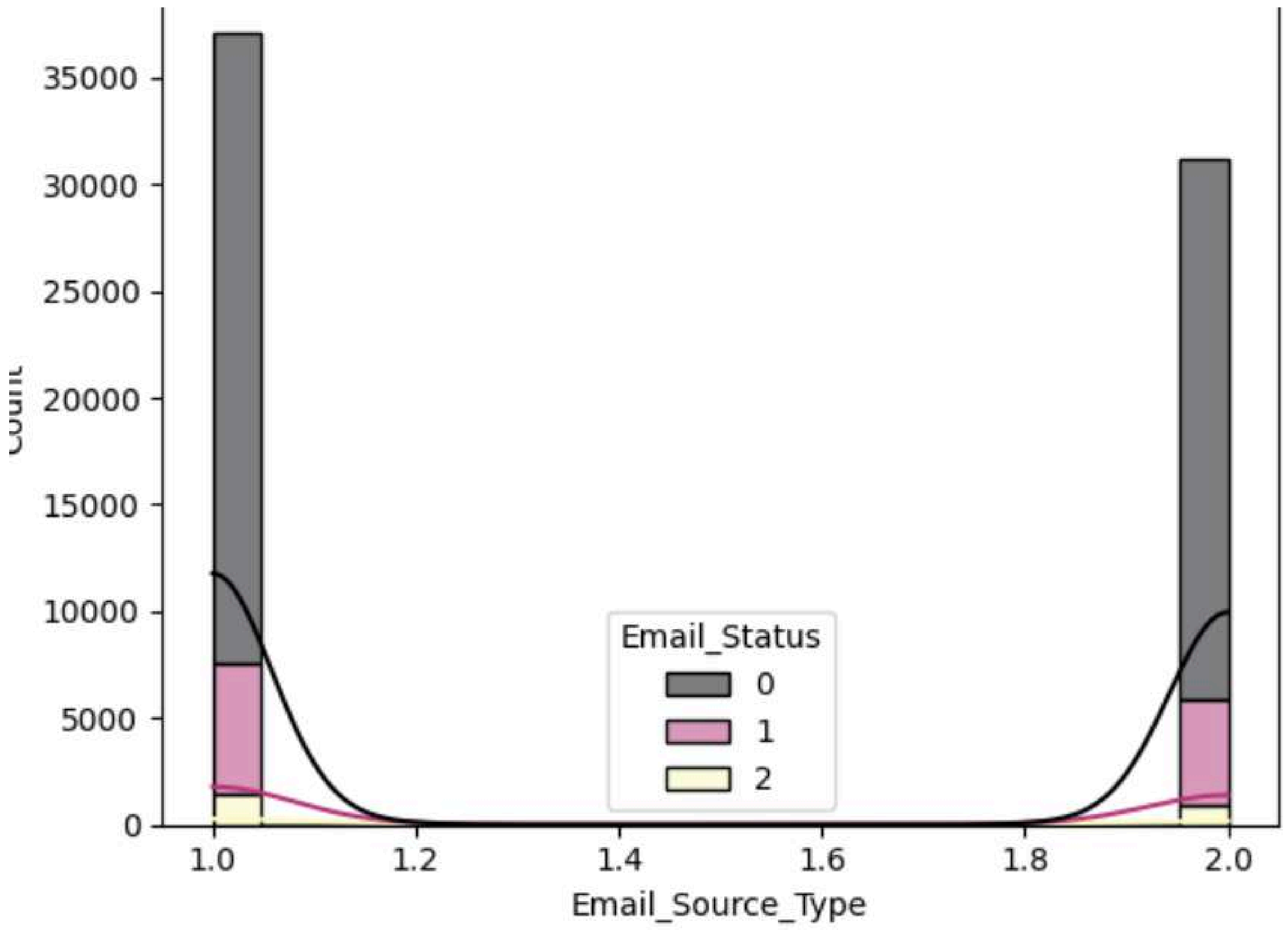
Email Campaign Strategy:
Companies may need to re-evaluate content and delivery strategies for both types of emails to increase open and response rates.

Subject Hotness Score



Based on graphic analysis, it can be seen that the majority of emails have a low Subject_Hotness_Score, in the range of 0 to 1. Of these emails, the majority are not opened, as shown by the high number of emails with a status of 0 (not opened) on this score. In contrast, the number of emails opened (state 1) and replied to (state 2) is much lower, and the frequency decreases as Subject_Hotness_Score increases. This shows that even a higher Subject_Hotness_Score does not guarantee that an email will be opened or replied to. In fact, most emails with low scores remain unopened. From this, it can be concluded that strategies for improving the quality of email subjects need to be paid attention to, because emails with a low Subject_Hotness_Score dominate and most of them are not opened. This suggests that efforts to increase recipient engagement through improving the quality of email subjects may be necessary.

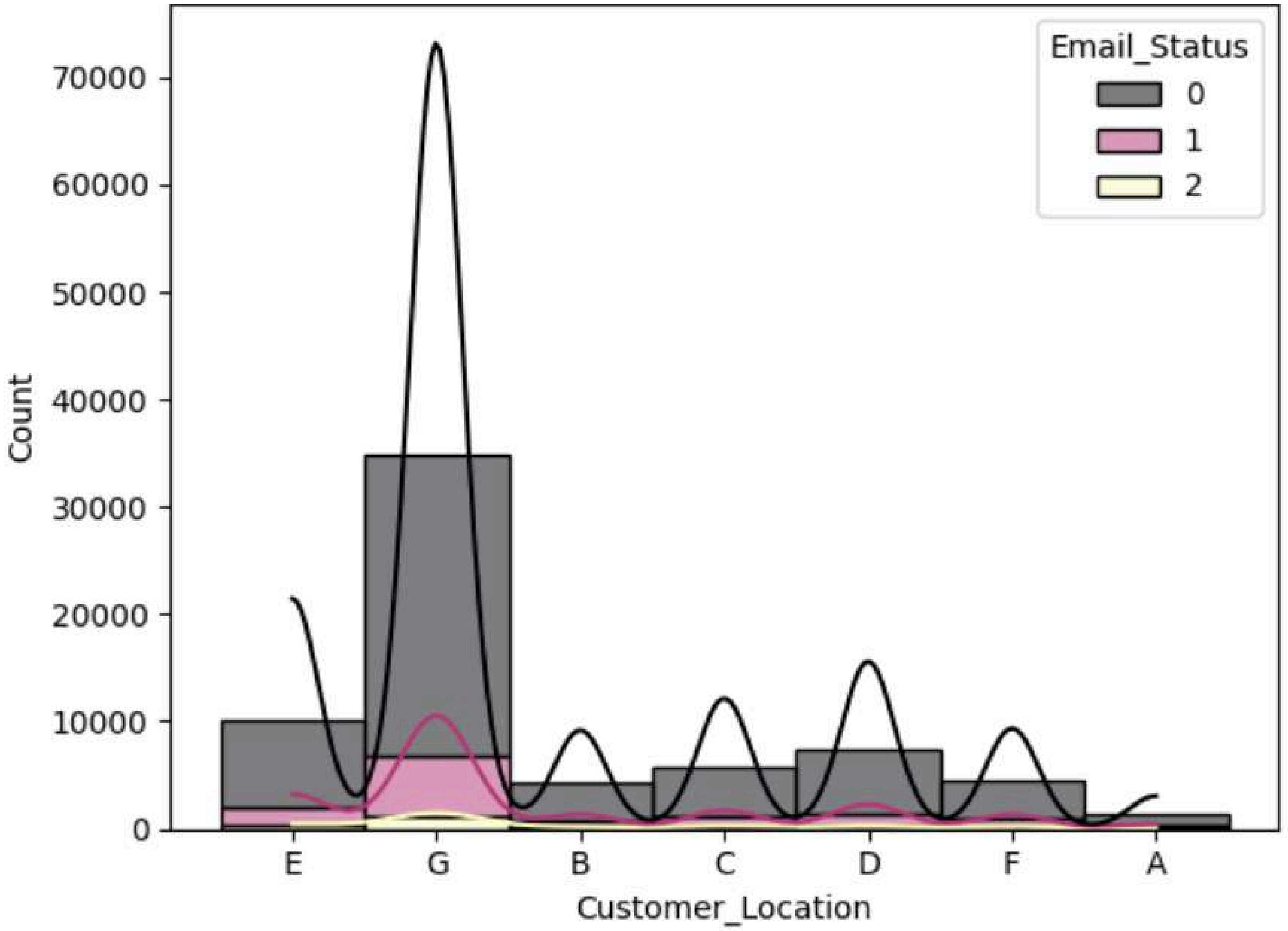
Email Source Type



Effectiveness of Email Sources: Both email sources (1 and 2) seem to be ineffective in grabbing recipients' attention, as most of them do not open the emails.

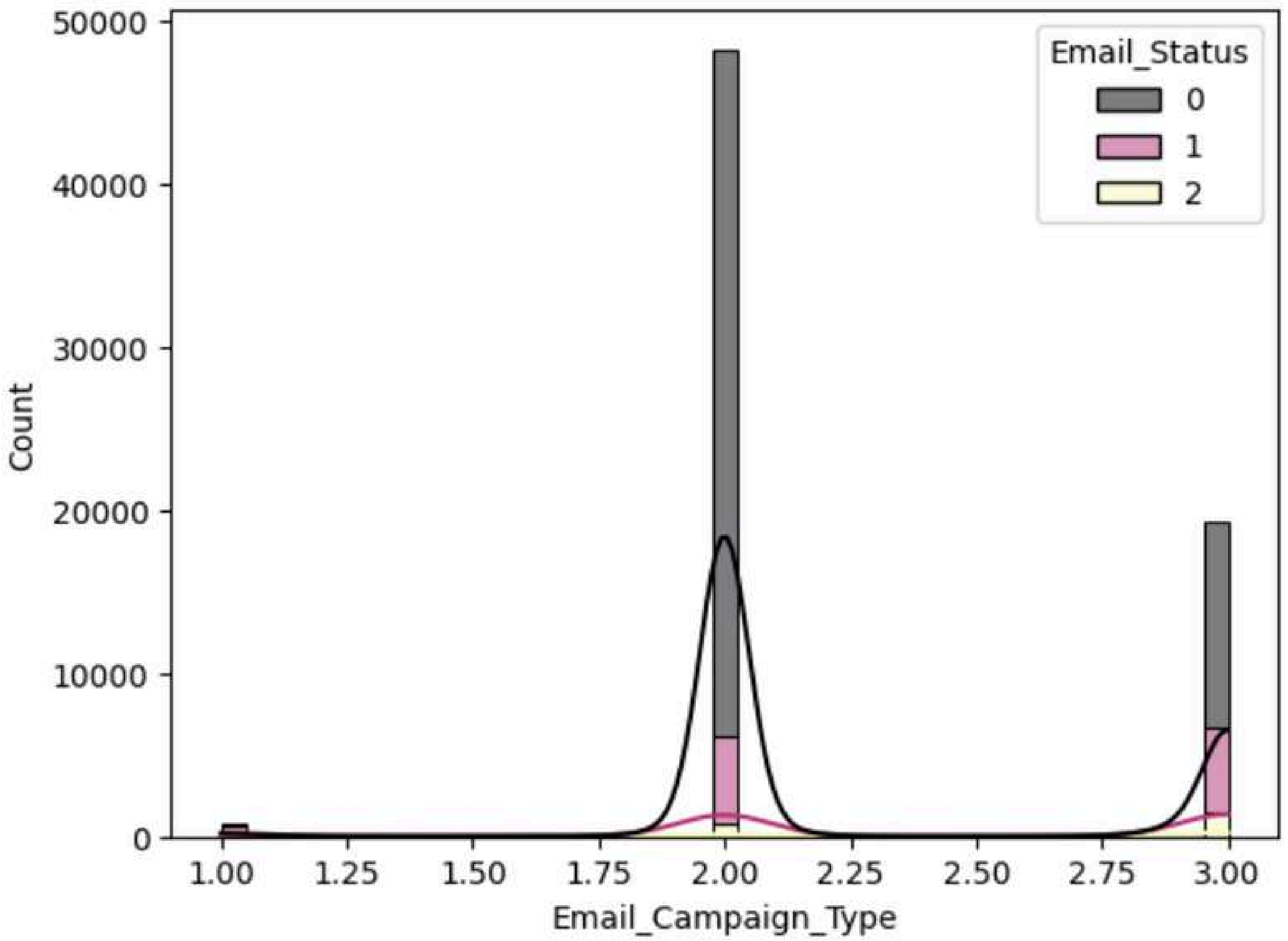
Email Campaign Strategy: Companies may need to re-evaluate sending strategies from these two sources to increase open and response rates.

Customer Location



Insights from this graph show that location G has a very dominant number of customers, indicating greater market potential at this location compared to other locations. Most customers have an email status of 0, which could mean they are inactive customers or have not responded to emails. Other locations have lower customer numbers, but locations E and D show less potential than locations B, C, F, and A. This suggests that a more intensive and focused marketing or communications strategy may be needed in these locations to increase engagement customer. Additionally, efforts to increase customer engagement with email states 1 and 2 are also necessary to ensure they stay connected and active.

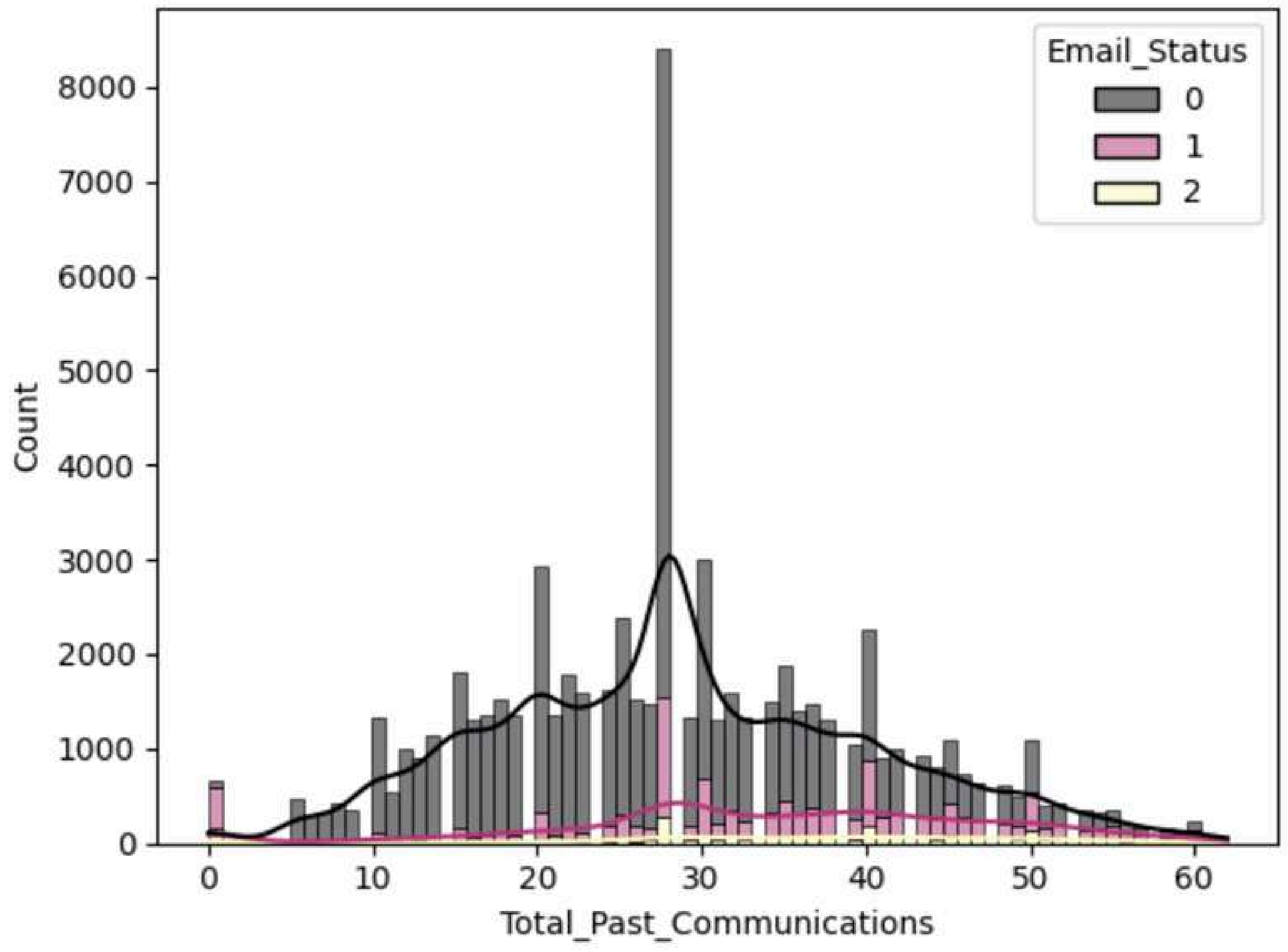
Email Campaign Type



The graph shows that email campaign type 2 has the highest number of subscribers, with the majority having an email status of 0, indicating that they are inactive or not engaged. Campaign type 3 has a significant but lower number of subscribers than type 2, while campaign type 1 has almost no engaged subscribers, indicating low effectiveness or lack of use of this campaign type.

It can be concluded that type 2 email campaigns need to be evaluated and improved to change email status 0 to a more active status. Campaign type 1 requires revisions or new strategies to increase customer engagement. Optimizing type 3 campaigns is also important to increase their effectiveness based on existing customer responses.

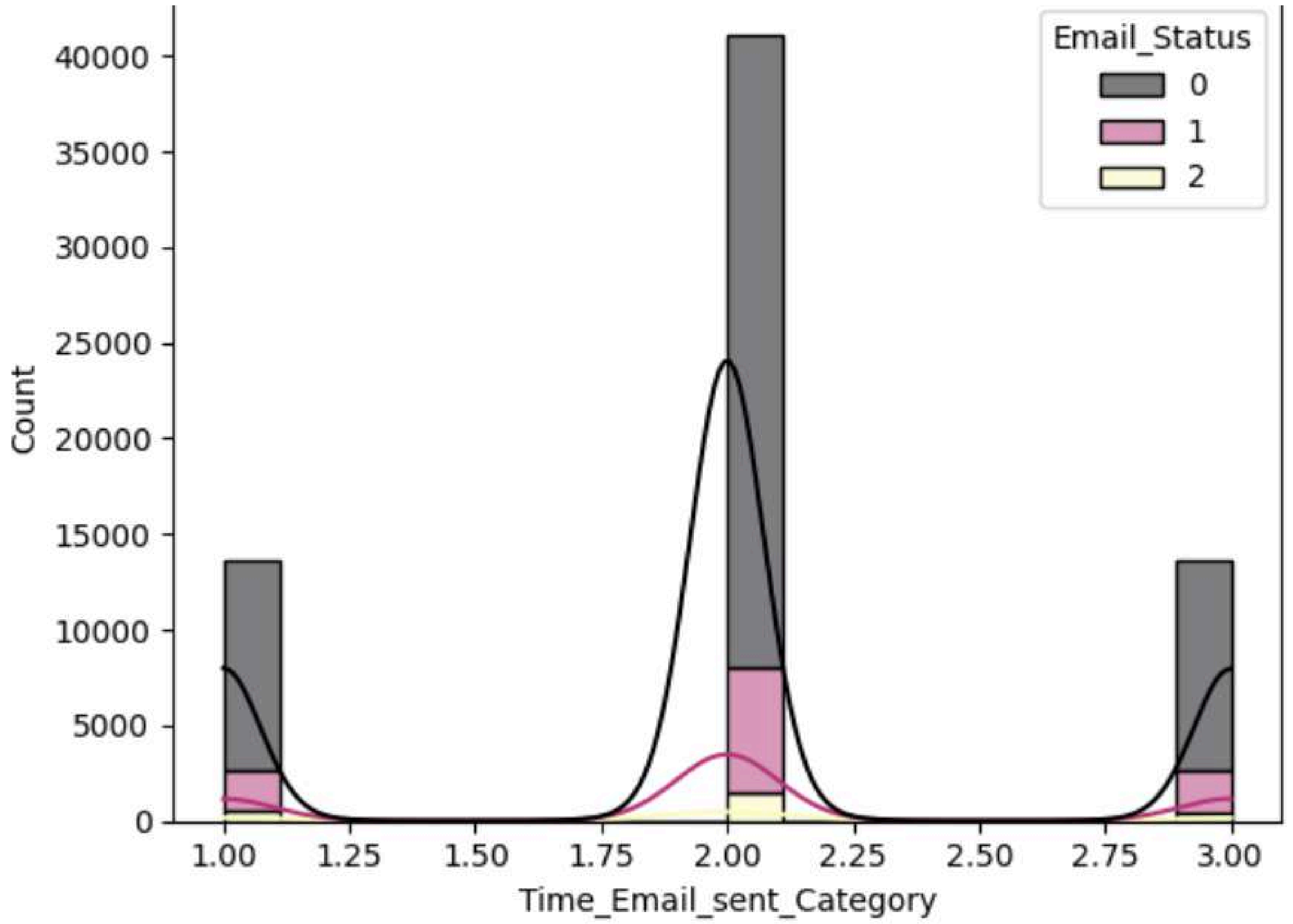
Total Past Communications



The graph shows that customers who have communicated approximately 30 times represent the highest peak in the distribution of the number of communications, with the majority of them having an email status of 0. This distribution also shows that there are customers who have communicated a varying number of times up to more than 50 times, indicating variations in engagement customer.

Insights from the second graph show that efforts need to be focused on customers who have communicated approximately 30 times to increase engagement and change their email status from 0 to more active. Customers with low communication volumes require a more personalized approach, while reactivation strategies are necessary for customers with very high communication volumes to keep them engaged.

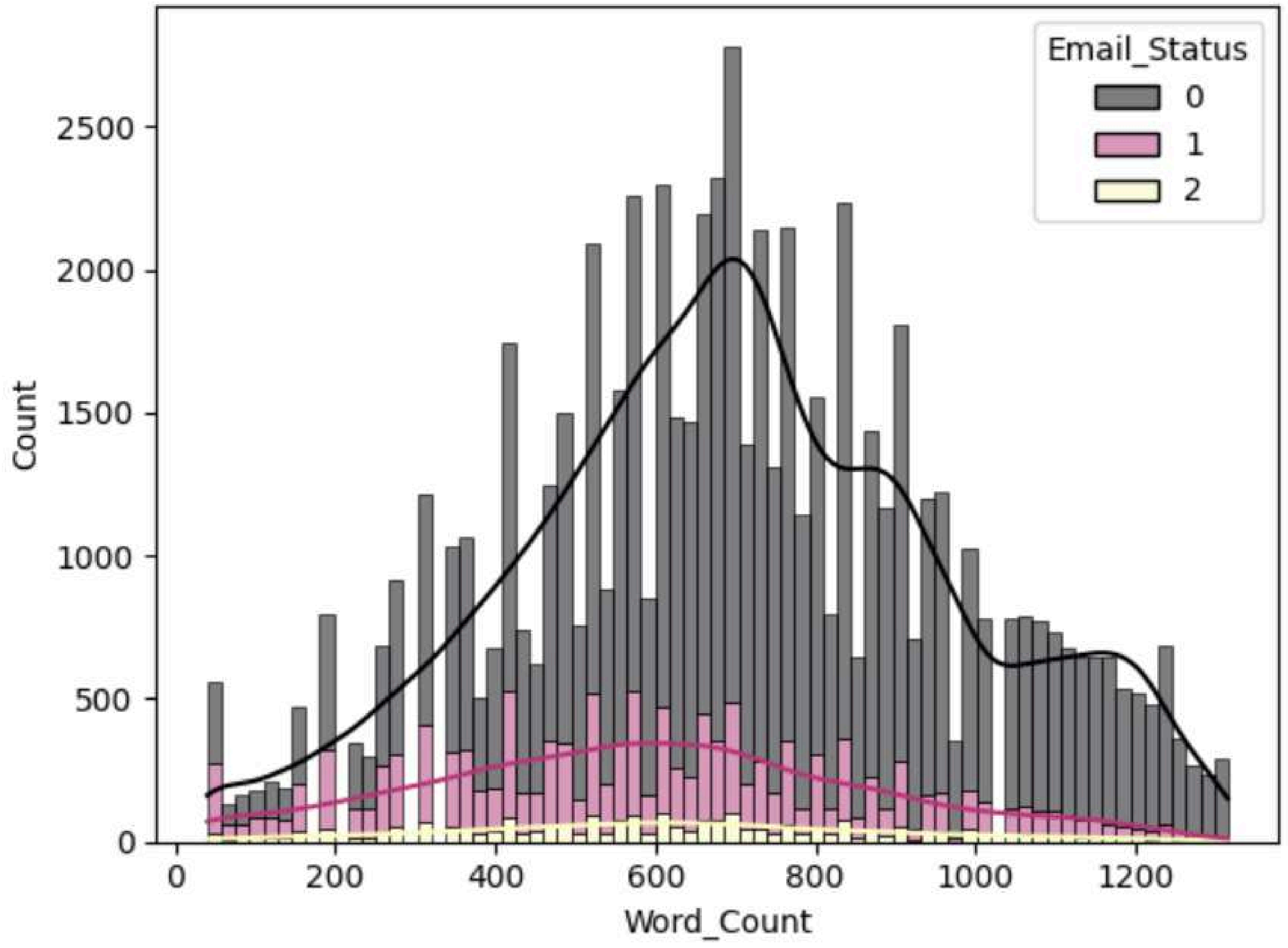
Time Email sent Category



From this graph, we can conclude that time category 2 has the highest number of emails sent, with the majority of emails status 0 (marked in gray). Time categories 1 and 3 have a smaller number of emails sent compared to category 2. Email status 0 dominates in all time categories, while statuses 1 and 2 only have a small number in all time categories.

Time category 2 has the highest number of emails sent, companies should consider sending more emails at this time. While time category 2 is dominant, companies should also consider testing the effectiveness of time categories 1 and 3 to see if there are opportunities to increase customer engagement at those times

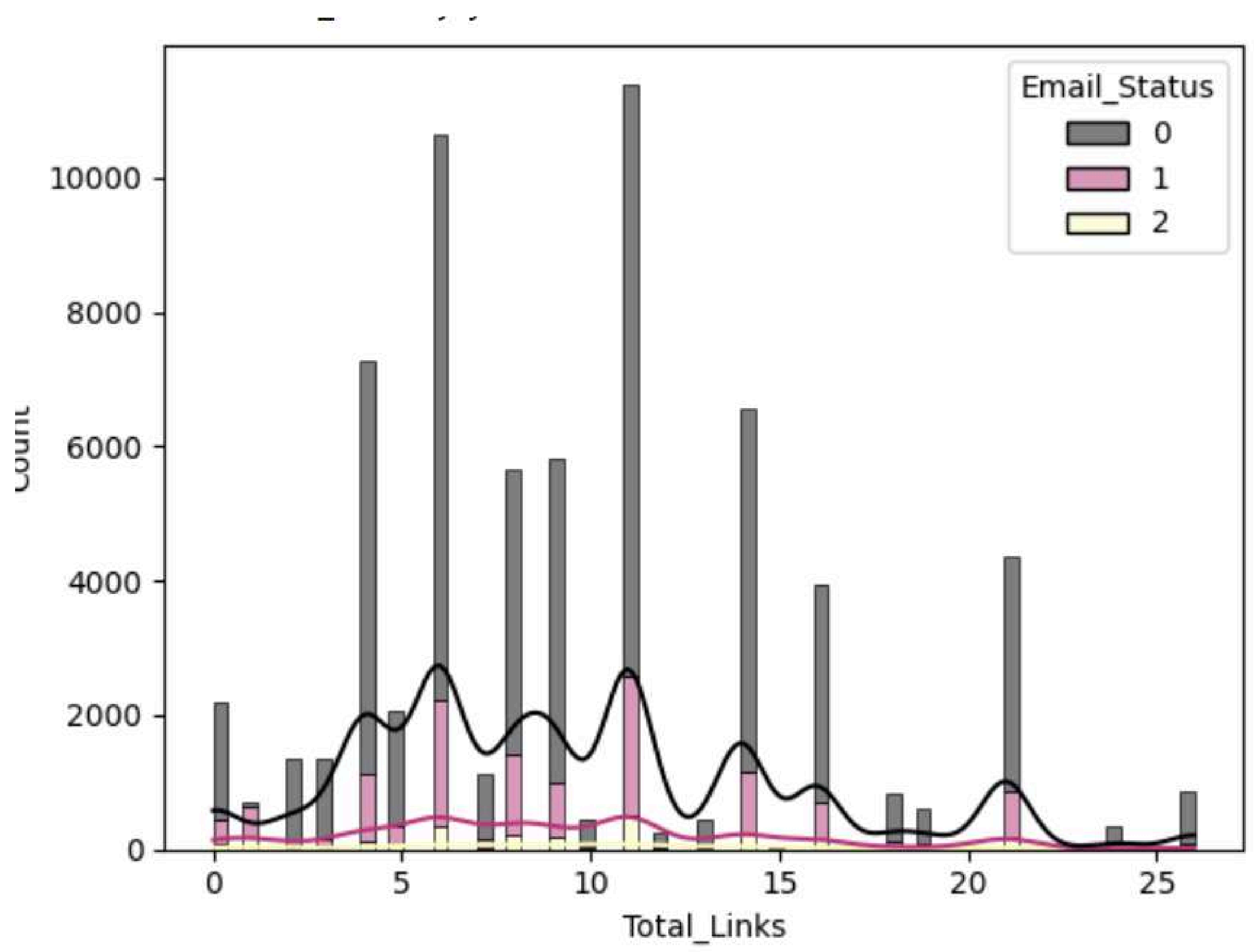
Word Count



From this graph, we can conclude that the majority of emails have a word count of around 500-800 words with an email status of 0. The word count distribution looks like a normal distribution with a peak around 500-600 words. Email status 0 (gray) dominates in all word count ranges, while emails with status 1 (pink) and 2 (yellow) have a smaller number compared to status 0 in all word count ranges.

With the majority of emails sent having a word count of around 500-800 words and email status 0 (unopened) dominating, companies should consider evaluating the content of the emails sent. It may be necessary to simplify the message, increase relevance, or make the email subject more interesting

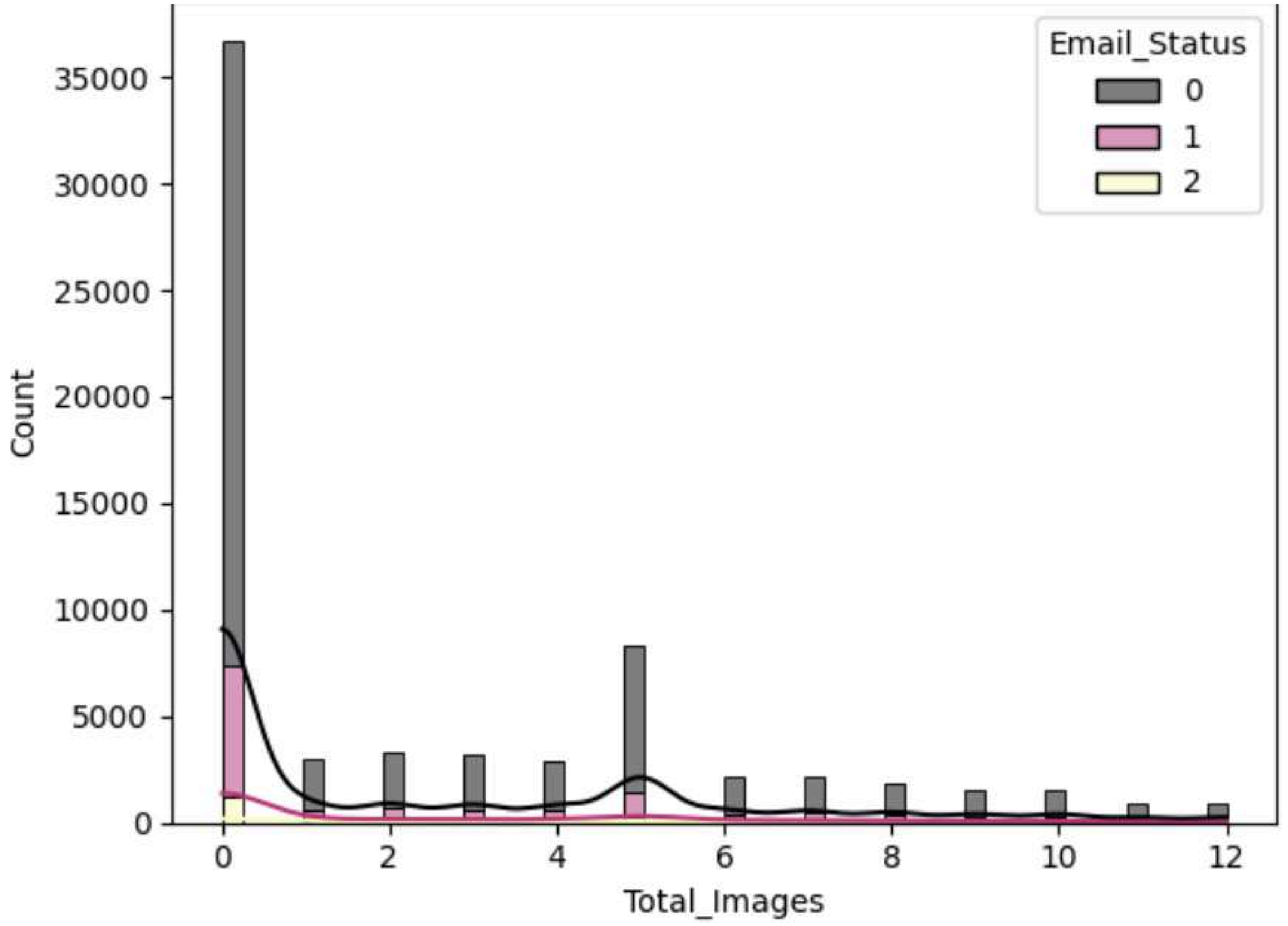
Total links



From this graph, we can conclude that most emails have a total of around 5-10 links with an email status of 0. The word count distribution looks like a normal distribution with a peak around 10 links. Email status 0 (gray) dominates in all word count ranges, while emails with status 1 (pink) and 2 (yellow) have a smaller number compared to status 0 in all link count ranges.

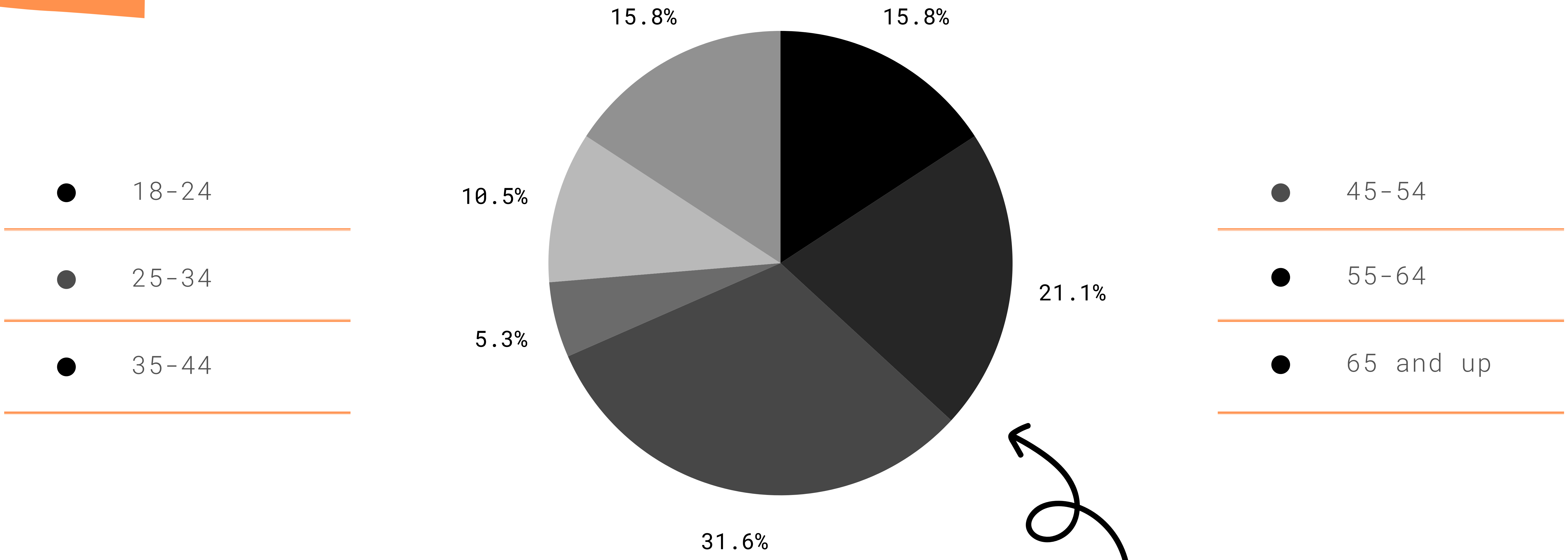
With the majority of emails sent getting email status 0 (not opened) dominating, companies should consider reducing the number of links in emails. Too many links can make an email look like spam. Maybe companies should not provide more than 15 links because we see that the total number of links above 15 is decreasing continuously. Companies can also gradually add links as needed.

Total Images



From this graph, we can conclude that the majority of emails have a total of 0 images with an email status of 0. Email status 0 (grey) dominates in all ranges of image counts, while emails with status 1 (pink) and 2 (yellow)) has a smaller number.

companies should evaluate the content of included images. Images should support the main message of the email and provide additional value for the recipient, such as showing featured products, special promotions, or interesting illustrations that support the text content. And also companies may not need to use images because we see that Total images 0 gets quite a lot of email statuses 1



CORRELATION

	Email_Status
Email_Type	-0.015074
Email_Status	1.000000

	Email_Status
Subject_Hotness_Score	-0.151514
Email_Status	1.000000

	Email_Status
Email_Source_Type	-0.024527
Email_Status	1.000000

	Email_Status
Customer_Location_Encoded	0.001459
Email_Status	1.000000

	Email_Status
Email_Campaign_Type	0.18551
Email_Status	1.00000

	Email_Status
Total_Past_Communications	0.231601
Email_Status	1.000000

	Email_Status
Time_Email_sent_Category	0.000051
Email_Status	1.000000

	Email_Status
Word_Count	-0.171116
Email_Status	1.000000

	Email_Status
Total_Links	-0.023717
Email_Status	1.000000

	Email_Status
Total_Images	-0.004439
Email_Status	1.000000

explanation

- **Variables that Have a Positive Correlation with Email_Status:**

Total_Past_Communications has the highest positive correlation with Email_Status (0.231601). This indicates that the more previous communications, the more likely the email status is active.
Email_Campaign_Type also has a positive correlation with Email_Status (0.18551). This shows that the type of email campaign can influence the status of the email.

- **Variables that Have a Negative Correlation with Email_Status:**

Subject_Hotness_Score has a negative correlation with Email_Status (-0.151514). This suggests that higher subject interest scores may correlate with inactive email status.
Word_Count has a negative correlation (-0.171116). This suggests that emails with a higher word count may be more likely to have an inactive status.

- **Variables that Have Almost No Correlation with Email_Status:**

Customer_Location_Encoded (0.001459), Time_Email_sent_Category (0.000051), Email_Source_Type (-0.024527), Total_Links (-0.023717), and Total_Images (-0.004439) have very low correlation with Email_Status. This indicates that these variables have little effect on the status of the email.

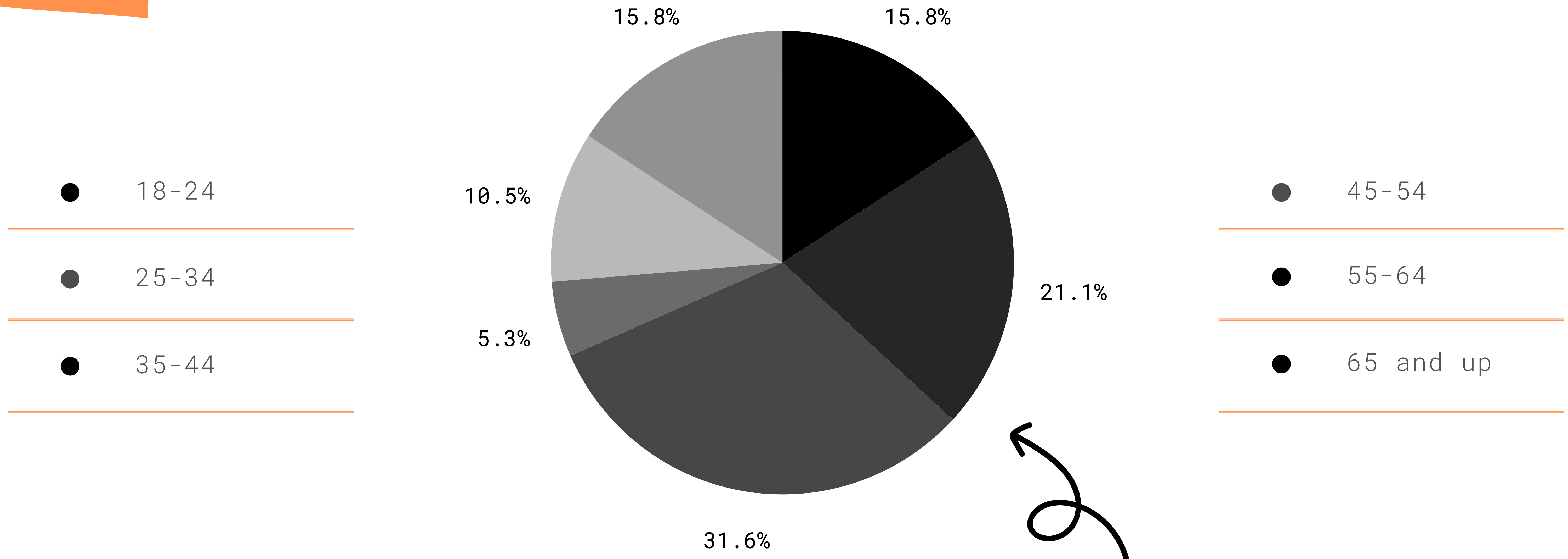
explanation

- **Actions for Variables with Low Correlation:**

Customer_Location_Encoded (0.001459), Time_Email_sent_Category (0.000051), Email_Source_Type (-0.024527), Total_Links (-0.023717), Total_Images (-0.004439):
Low correlation indicates that these variables do not have a significant influence on Email_Status.
Action: Even though these variables have low correlation, it is still important to monitor them for possible changes in the future. However, the main focus should be on variables with more significant correlations

- **Additional recommendations**

Customer Segmentation:
Segment customers based on their communication history and the types of campaigns they respond to. Personalize communications based on these segments to increase engagement.
Test A/B Testing:
Conduct A/B testing for email subjects and content length. Find the combination that is most effective in maintaining an active email status.



MODELING DATA

Logistic Regressiom

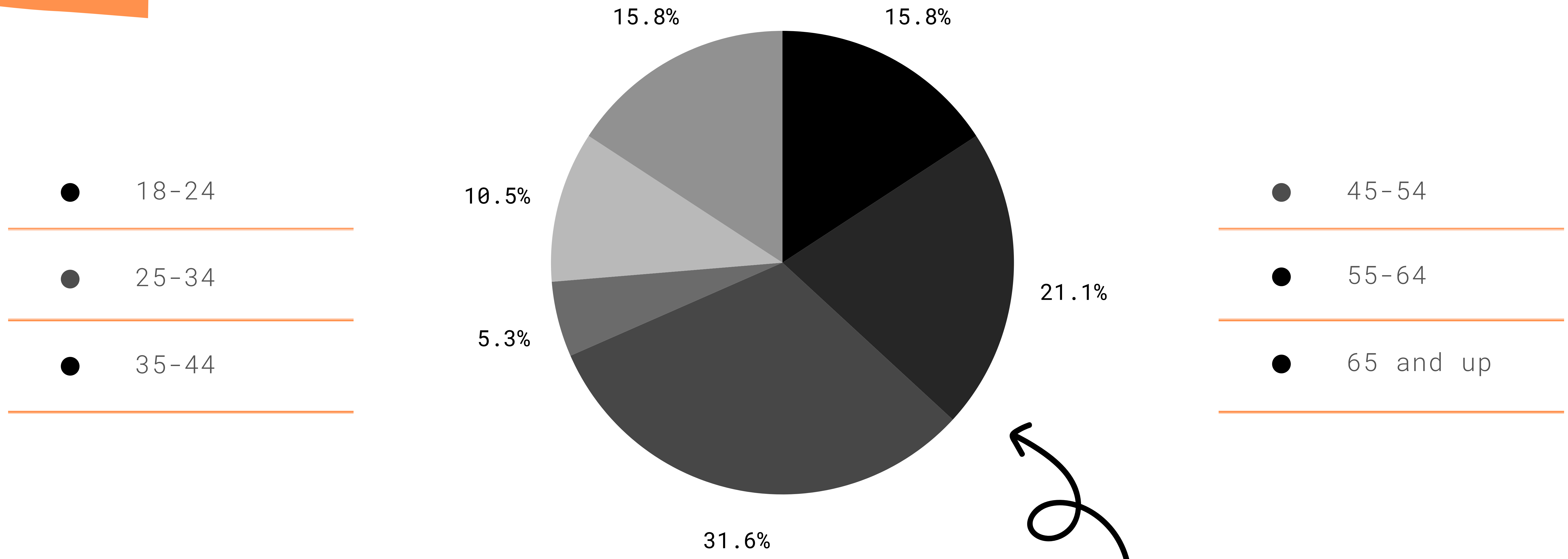
	precision	recall	f1-score	support
0	0.83	0.98	0.90	16505
1	0.48	0.14	0.22	3312
2	0.00	0.00	0.00	689
accuracy			0.81	20506
macro avg	0.43	0.37	0.37	20506
weighted avg	0.74	0.81	0.76	20506

Accuracy: 0.8094216326928704

Random forest

	precision	recall	f1-score	support
0	0.83	0.96	0.89	16505
1	0.42	0.17	0.24	3312
2	0.17	0.03	0.06	689
accuracy			0.80	20506
macro avg	0.47	0.39	0.40	20506
weighted avg	0.74	0.80	0.76	20506

Accuracy: 0.8010338437530479



CONCLUSION & RECOMMENDATION

CONCLUSION

Analysis shows that email engagement rates are low, with the majority of emails going unopened, especially for types 1 and 2. While a high Subject_Hotness_Score may seem promising, it doesn't always translate into higher open or reply rates. Type 2 campaigns have potential but many emails remain unopened, while type 1 campaigns show low effectiveness. Delivery time in category 2 shows the best results, but still needs further evaluation. Email content, such as word count and links, also affects openness, with too many words or links reducing effectiveness. Variables such as Total_Past_Communications show a positive correlation with active email status, while Subject_Hotness_Score and Word_Count show a negative correlation.

Recommendation

To increase the effectiveness of email campaigns, companies should focus on improving email content by creating more interesting and relevant subjects and simplifying messages. The number of words, links and images in emails also needs to be considered, by reducing elements that do not support the main message. Email sending strategies should be further evaluated, including testing sending times to find optimal times and adjusting sending frequency so as not to overwhelm recipients. Campaign types that are less effective should be improved, while campaign types that show potential need to be optimized. A focus on customer segmentation and a personalized approach can increase engagement, as well as the development of reactivation programs for inactive customers. Lastly, continuous monitoring of performance and adoption of a data-driven approach will help in adjusting strategies and improving email campaign results.

VIEW

COLLAB

GITHUB

THANK YOU!

WE'RE OPEN TO QUESTIONS AND CLARIFICATIONS.